

Evolving a Neural Network Agent for Planet Wars

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COM407: Computational Intelligence

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Roadmap

1. Planet Wars
2. Methodology
3. Results
4. Live Demo of the Best Agent
5. Q&A

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Planet Wars

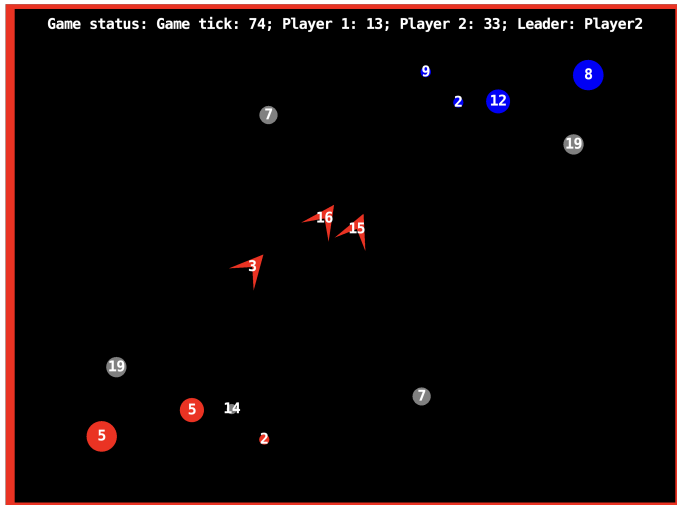


Figure: Example game state from the Planet Wars game.

Objectives of Planet Wars

- Control new planets to expand further and have wider resource production
- Use each planet's growth rate to accumulate ships and have strategic advantage
- Protect owned planets and balance offense/defense to not lose them while expanding
- Conquer all the planets or own more planets & resources until end of the time limit

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Features per Planet

- **Planet ownership** Encoded as +1 for us, -1 for opponent, and 0 for neutral
- **Ship count** Current number of ships on the planet, normalized by 200
- **Growth rate** The growth rate of the planet, normalized by the maximum growth rate
- **Planet position** Absolute coordinates normalized by map width and height
- **Incoming friendly ships** Total number of incoming friendly ships to this planet
- **Incoming enemy ships** Total number of incoming opponent ships to this planet
- **Transporter position** Position of the outgoing transporter, if none (0, 0)
- **Transporter velocity** Transporter velocity normalized by transporter speed, if none 0

Encoding the Game State

$$\mathbf{M} = \begin{bmatrix} o_{p_1} & c_{p_1} & \gamma_{p_1} & x_{p_1} & y_{p_1} & f_{p_1} & e_{p_1} & s_{p_1}^x & s_{p_1}^y & v_{p_1}^x & v_{p_1}^y \\ o_{p_2} & c_{p_2} & \gamma_{p_2} & x_{p_2} & y_{p_2} & f_{p_2} & e_{p_2} & s_{p_2}^x & s_{p_2}^y & v_{p_2}^x & v_{p_2}^y \\ o_{p_3} & c_{p_3} & \gamma_{p_3} & x_{p_3} & y_{p_3} & f_{p_3} & e_{p_3} & s_{p_3}^x & s_{p_3}^y & v_{p_3}^x & v_{p_3}^y \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ o_{p_{10}} & c_{p_{10}} & \gamma_{p_{10}} & x_{p_{10}} & y_{p_{10}} & f_{p_{10}} & e_{p_{10}} & s_{p_{10}}^x & s_{p_{10}}^y & v_{p_{10}}^x & v_{p_{10}}^y \\ o_{p_{11}} & c_{p_{11}} & \gamma_{p_{11}} & x_{p_{11}} & y_{p_{11}} & f_{p_{11}} & e_{p_{11}} & s_{p_{11}}^x & s_{p_{11}}^y & v_{p_{11}}^x & v_{p_{11}}^y \\ o_{p_{12}} & c_{p_{12}} & \gamma_{p_{12}} & x_{p_{12}} & y_{p_{12}} & f_{p_{12}} & e_{p_{12}} & s_{p_{12}}^x & s_{p_{12}}^y & v_{p_{12}}^x & v_{p_{12}}^y \end{bmatrix}$$

Figure: Representation of the complete game state

Neural Network Architecture

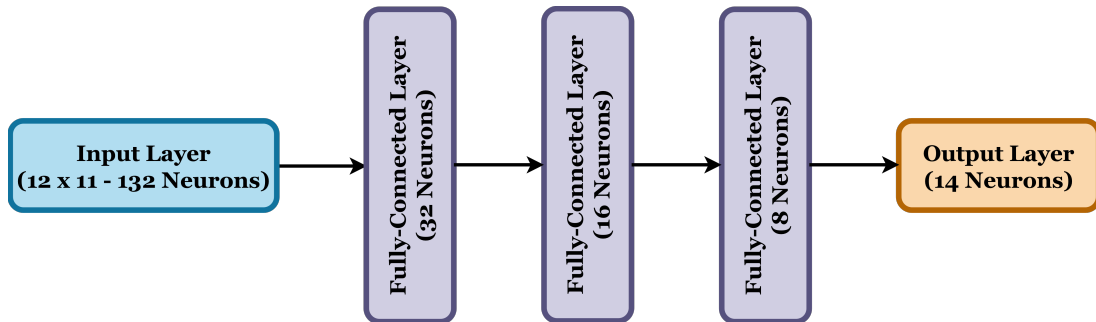


Figure: Architecture of the neural network that was used with 3 hidden layers of size 32, 16, and 8

Evolution via Covariance Matrix Adaptation Evolution Strategy

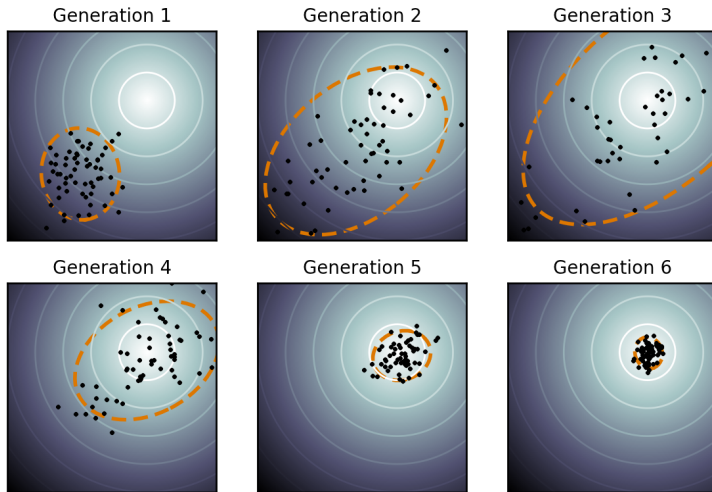


Figure: Visualization of the updates in the search space throughout the CMA-ES optimization

Training Setup

- The agent utilizes a neural network with 3 hidden layers with sizes 32, 16, and 8
- The agent is evolved via CMA-ES for 500 generations where the initial σ is 0.5
- Each candidate solution is evaluated 100 games against the greedy agent and the fitness value is the ratio of the games won
- Agent with the best fitness is selected at the end of 500 generations

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Baseline Agents

- **Pure Random Agent** Randomly picks the source and destination planet from *all* planets, ignoring ownership, ship counts, and context, attempts to send half of the ships from the source. Can attempt invalid moves.
- **Careful Random Agent** Randomly picks one of the owned idle planets and attacks a random opponent planet, always sending the half of the ships from the source. No invalid moves.
- **Greedy Heuristic Agent** Chooses the strongest planet it owns and attacks to a planet considering the enemy ship count, distance, and growth rate in the target planet. Sends ships only when the attack is likely to succeed.

Benchmark of the Baseline Agents

Table: Pairwise win rates of the baseline agents in 100,000 games where the row agent wins against the column agent.

	Pure Random	Careful Random	Greedy Heuristic
Pure Random	–	5.76%	3.05%
Careful Random	94.24%	–	37.12%
Greedy Heuristic	96.95%	62.88%	–

Fitness Progression of the Neural Agent

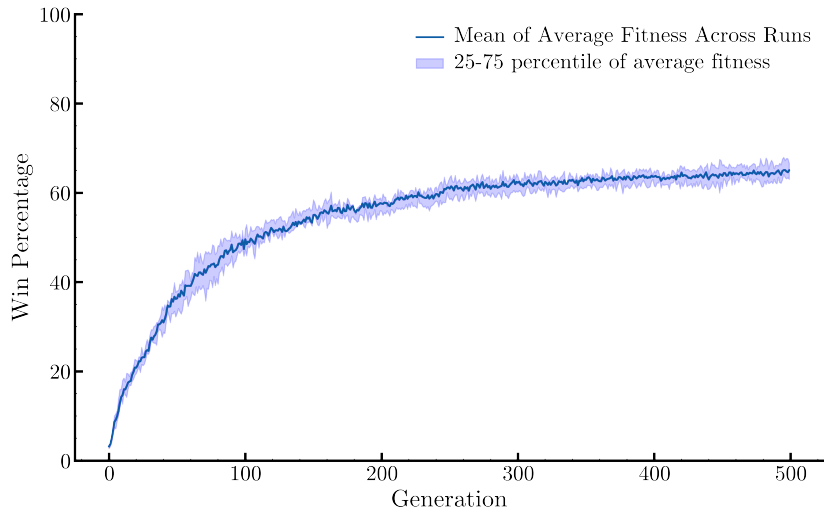


Figure: Mean and 25th – 75th percentile fitness progression for 10 independent training runs

Neural Agent Against the Baseline Agents

Table: Performance metrics of the best neural agent against the greedy heuristic agent over 100,000 games

Metric	Neural Agent	Greedy Heuristic
Win Rate	61.08%	38.92%
Avg. Planets Owned	4.432 ($\sigma = 3.118$)	5.502 ($\sigma = 4.543$)
Avg. Ships Owned	246.941 ($\sigma = 264.622$)	190.401 ($\sigma = 240.199$)
Avg. Neutral Planets	2.065 ($\sigma = 1.927$)	

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